# DATA PREPROCESSING REPORT

## House Price Prediction Using Machine Learning

**Dataset:** Ames Housing Dataset
**Project Type:** Regression Analysis
**Tools & Technologies:** Python, Jupyter Notebook, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn

# 1. Introduction

## 1.1 Project Overview

House price prediction is one of the most common regression problems in machine learning. The objective of this project is to predict the selling price of residential houses using historical housing data from the Ames Housing dataset.

Before training machine learning models, the raw dataset must be transformed into a clean and structured format. Data preprocessing is a crucial step because the quality of the input data directly affects the performance of prediction models.

This report explains the preprocessing techniques applied to prepare the dataset for machine learning.

## 1.2 Objectives

The main objectives of data preprocessing are:

- To understand the dataset structure.
- To identify missing values and duplicate records.
- To clean and transform the dataset.

- To engineer meaningful features.
- To prepare the dataset for machine learning algorithms.

# 2. Dataset Overview

The project uses the **Ames Housing Dataset**, which contains detailed information about residential houses sold in Ames, Iowa.

## Dataset Information

| Attribute | Value |
|---|---|
| Dataset Name | Ames Housing Dataset |
| Total Records | 1460 |
| Problem Type | Regression |
| Target Variable | SalePrice |
| Data Type | Numerical & Categorical |
| Programming Language | Python |

## Dataset Features

The dataset contains several property-related features, including:

- Lot Area
- Overall Quality
- Overall Condition
- Ground Living Area
- Basement Area
- Garage Capacity
- Year Built
- Number of Bathrooms
- Sale Price

These features were used throughout the machine learning pipeline.

# 3. Data Quality Assessment

Before applying preprocessing techniques, the dataset was thoroughly analyzed to identify potential quality issues.

The following analyses were performed:

- Dataset shape inspection
- Data type verification
- Missing value analysis
- Duplicate record detection
- Statistical summary generation
- Target variable inspection

These checks ensured that the dataset was suitable for further analysis and model development.

## Data Quality Checklist

| Validation | Status |
|---|---|
| Dataset Loaded Successfully | completed |
| Data Types Verified | completed |
| Missing Values Identified | completed |
| Duplicate Records Checked | completed |
| Statistical Summary Generated | completed |

# 4. Missing Value Handling

Some features contained missing values that required appropriate treatment before model training.

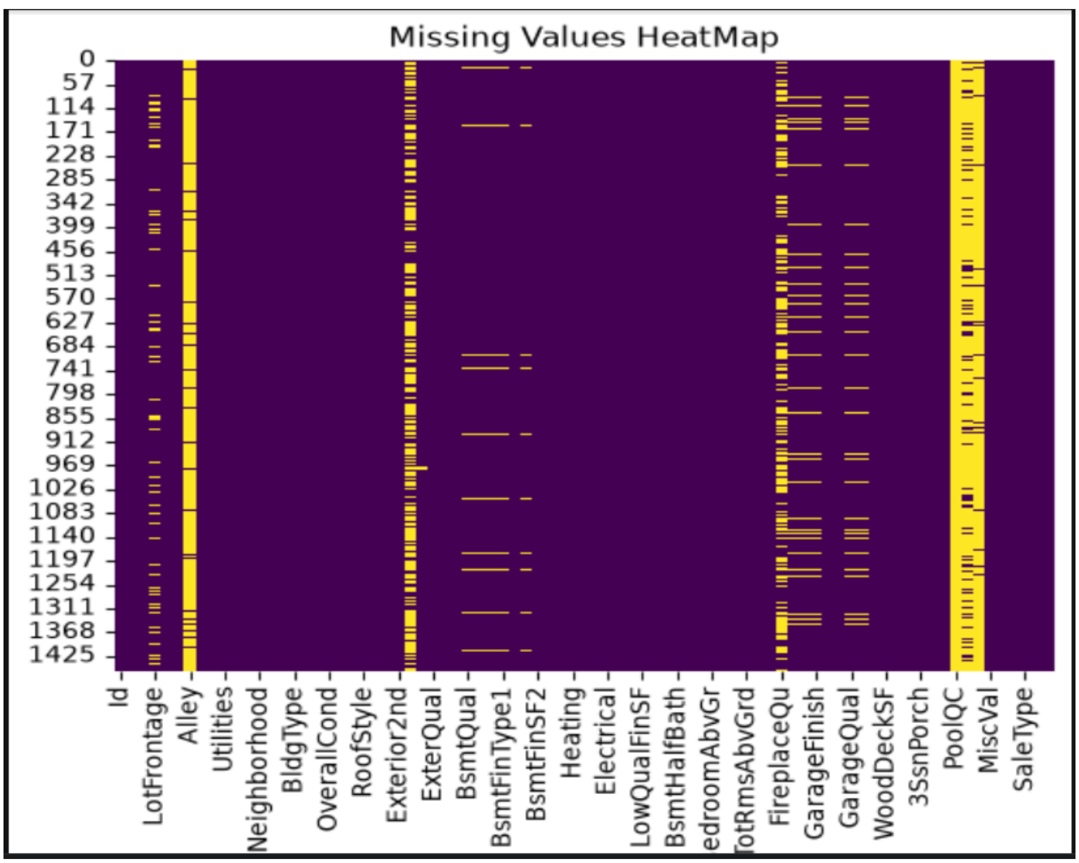
Different strategies were applied depending on the amount and type of missing data.

## Missing Value Treatment

| Feature | Treatment Method |
|---|---|
| LotFrontage | Median Imputation |
| GarageYrBlt | Median Imputation |
| MasVnrArea | Median Imputation |
| PoolQC | Removed (High Missing Percentage) |
| Alley | Removed (High Missing Percentage) |
| Fence | Removed (High Missing Percentage) |
| FireplaceQu | Appropriate Imputation |
| Garage Features | Appropriate Imputation |

## Observation

Features with extremely high missing percentages were removed because they provided limited useful information. Numerical features with relatively few missing values were filled using the median to preserve the distribution of the data.



# 5. Feature Engineering

To improve prediction accuracy, additional features were created from existing variables.

## Engineered Features

| New Feature | Formula | Purpose |
|---|---|---|
| TotalHouseArea | Basement + First Floor + Second Floor | Represents total usable house area |
| TotalBathrooms | Full + Half + Basement Bathrooms | Represents total bathrooms |
| HouseAge | YrSold – YearBuilt | Indicates house age |
| RemodelAge | YrSold – YearRemodAdd | Indicates years since renovation |
| TotalPorchSF | Sum of Porch Areas | Represents total outdoor space |

## Benefits

Feature engineering improves the dataset by combining related information into meaningful variables. These new features provide additional insights and help machine learning models make more accurate predictions.

# 6. Data Encoding

The dataset contained several categorical variables that could not be directly processed by regression algorithms.

These categorical features were converted into numerical values using encoding techniques, making the dataset compatible with machine learning models.

### Advantages of Encoding

- Converts categorical data into numerical format.
- Enables regression models to process all features.
- Improves model compatibility.
- Preserves useful information from categorical variables.

# 7. Final Dataset Summary

After completing all preprocessing tasks, the dataset became clean and suitable for model development.

### Final Summary

| Process | Status |
|---|---|
| Missing Values | Completed |
| Duplicate Check | Completed |
| Feature Engineering | Completed |
| Data Encoding | Completed |
| Dataset Ready for Modeling | Yes |

# 8. Key Outcomes

The preprocessing stage successfully transformed the raw housing dataset into a structured and machine learning-ready dataset.

### Major Achievements

- Dataset cleaned successfully.
- Missing values handled appropriately.
- Unnecessary features removed.
- New informative features created.
- Categorical variables encoded.
- Dataset prepared for regression modeling.

# 9. Conclusion

Data preprocessing is one of the most important stages in a machine learning project. In this project, the Ames Housing dataset was successfully cleaned and transformed through missing value handling, feature engineering, and data encoding.

The preprocessing process improved the quality of the dataset and ensured that it was ready for training and evaluating multiple regression models in the next phase of the project.